

Fault-Tolerant Swarm Quadrotor Control for Non-Convex Geodetic Mapping

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Abstract. This paper presents a three-layer control architecture for autonomous geodetic mapping of non-convex regions by a quadrotor swarm. The high-level layer partitions the region by centroidal Voronoi tessellation, generates a polygon-aware boustrophedon sweep for each cell, and redistributes coverage through Fault-Tolerant Cooperative Coverage (FT-CC) on agent loss. The mid-level layer enforces safety through a control barrier function quadratic program (CBF-QP) solved once per agent per control tick, with a class- $\mathcal{K}$ function derived from the vehicle tilt limit and dead time. The low-level layer is a cascaded PID whose gains are synthesised either from the algebraic Riccati equation (LQR) or from the H_∞ Riccati equation. The architecture is validated in ROS 2 and Gazebo over three non-convex regions and two controllers along a load ladder that adds one stressor at a time: nominal, turbulence, turbulence with obstacles, and turbulence with obstacles and sequential agent loss. Coverage held at **XX.X-XX.X%** with zero inter-agent collisions throughout; H_∞ reduced cross-track RMS error by **XX%** under turbulence at **X.Xx** the control effort, and FT-CC recovered coverage to within **XX** percentage points of the no-failure condition after losing two of seven agents.

INTRODUCTION

Swarm quadrotor systems support autonomous geodetic mapping in disaster response, precision agriculture, environmental surveillance, and infrastructure inspection. Partitioning a territory among several agents converts a sequential survey into a parallel one, and the saving grows with the size of the area.

The demands this places on the system are easiest to see in a concrete case. Consider mapping a landslide scar a few hundred metres across, a few days after the event. The boundary is irregular and encloses ground that must not be overflown, so the survey region is non-convex. Debris and standing structures are not on any map, so obstacles must be found in flight instead of looked up. Gusts channel through the terrain, and a quadrotor that drifts off its sweep line leaves a gap in the map, not merely a tracking error. If one vehicle is lost, the area it had been assigned is still unsurveyed, and the remaining vehicles have to absorb it before the light goes. A system that handles only the first of these is of little use in the field.

Meeting all four demands at once therefore requires region partitioning that tolerates non-convex geometry, obstacle handling that does not presume prior knowledge, flight control that holds a line under disturbance, and coverage that survives the loss of an agent.

Four gaps persist across the literature reviewed in the next section. First, most coverage algorithms assume a *convex* operational region [1, 2], whereas survey areas are non-convex by construction. Second, end-to-end pipelines from partitioning down to rotor control are scarce [2]: planning and execution are normally developed and reported separately, hiding the integration mismatches that appear only when they are connected. Third, fault tolerance under actuator or communication failure is underexplored [3], although the probability that *some* agent fails grows with swarm size. Fourth, validation is predominantly carried out on kinematic point-mass models, not on full quadrotor dynamics [4]; path lengths, mission durations, and safety margins obtained that way are measured on a vehicle whose tilt limit, motor lag, and dead time do not exist.

This paper makes four contributions.

- (i) A polygon-aware coverage planner whose extremal sweep rows follow the cell boundary. This removes the unswept wedges that a fixed-pitch sweep leaves at tapered corners.
- (ii) A CBF-QP that folds static obstacles, moving obstacles, arena walls, and reciprocal inter-agent spacing into a single quadratic program. Its class- $\mathcal{K}$ function is derived from the identified closed-loop plant, so every admissible command is one the vehicle can execute.

- (iii) A like-for-like comparison of LQR and H_∞ gain synthesis on an identical cascade, including an explicit procedure for selecting the attenuation bound γ .
- (iv) An evaluation on full six-degree-of-freedom dynamics over three non-convex regions, under nominal, turbulent, and agent-failure conditions.

The remainder of the paper surveys the related literature, describes the three layers and the simulation environment, and reports the measured results.

RELATED WORK

Coverage Control and Region Partitioning

Coverage control by centroidal Voronoi tessellation was established by Cortés *et al.* [5], who showed that a Lloyd descent [6] on the Voronoi centroids [7] converges to a locally optimal sensor configuration. The formulation is elegant but assumes a convex domain: the centroid of a non-convex cell may fall outside the cell, and the geodesic and Euclidean metrics diverge. Extensions to non-convex domains take three broad routes — diffeomorphic mapping to a convex proxy, geodesic distance computation, and local navigation around obstacles. Breitenmoser *et al.* [8] combine Lloyd descent with the TangentBug local planner and prove convergence in non-convex polygons; Lee and Lee [9] instead add agent dropout and reinsertion to escape the local minima that non-convex domains introduce; Cao *et al.* [10] use buffered Voronoi cells to embed collision margins directly in the partition. The present work follows the third route in spirit but keeps the partition purely geometric: cells are clipped against the survey polygon itself, and safety margins are enforced downstream by the barrier layer rather than baked into the cell boundaries.

Coverage Path Planning

Given a cell, the sweep itself is a coverage path planning problem, surveyed by Choset [11] and later by Galceran and Carreras [12]. Boustrophedon cellular decomposition [13] remains the standard: the free space is decomposed into cells sweepable by simple back-and-forth motions. Most implementations sweep with a fixed pitch and a fixed direction, which is exact for rectangular cells and leaves systematic gaps at tapered corners — precisely the geometry that non-convex Voronoi partitioning produces most often. Route optimisation in this family typically targets the number of turns and the total path length [2, 12], and is normally evaluated on kinematic agents, so the turn cost is a count and carries no dynamic penalty.

Reciprocal Collision Avoidance and Safety Filters

For multi-agent deconfliction, control barrier functions provide forward invariance of a designated safe set through a single inequality constraint on the control [14, 15]. Wang *et al.* [16] extend this to reciprocal multi-robot safety with safety barrier certificates, and Zhou *et al.* [17] achieve a comparable guarantee geometrically through buffered Voronoi cells. In the canonical formulation the class- $\mathcal{K}$ function is linear, $\alpha(h) = \gamma h$, with γ chosen by tuning. That choice is adequate when the actuator can deliver the implied deceleration, but a quadrotor’s lateral acceleration is bounded by its tilt limit and its command path carries dead time, so the admissible approach speed near contact is dictated by braking capability, not by a tuning constant. The barrier used here, Eq. (17), is derived from that braking capability directly.

Fault Tolerance in Swarms

Fault tolerance in robot swarms is predominantly reactive: a failure is detected, then the task is redistributed. O’Keeffe [3] argues for the predictive alternative, resolving degradation before it becomes failure, and reports comparable or better performance than a reactive baseline. The mechanism used here is reactive by design — heartbeat timeout, compound-polygon merge, helper reallocation — because the failures of interest are abrupt losses rather than gradual degradation, but the two approaches are complementary.

METHODOLOGY

System Architecture

Figure 1 shows the three layers. The high-level layer consumes the survey polygon and initial poses and emits waypoints. The mid-level layer maps the desired velocity to a certified-safe commanded velocity. The low-level layer tracks the resulting reference and outputs rotor speeds. Odometry closes the loop at every level. The state machine

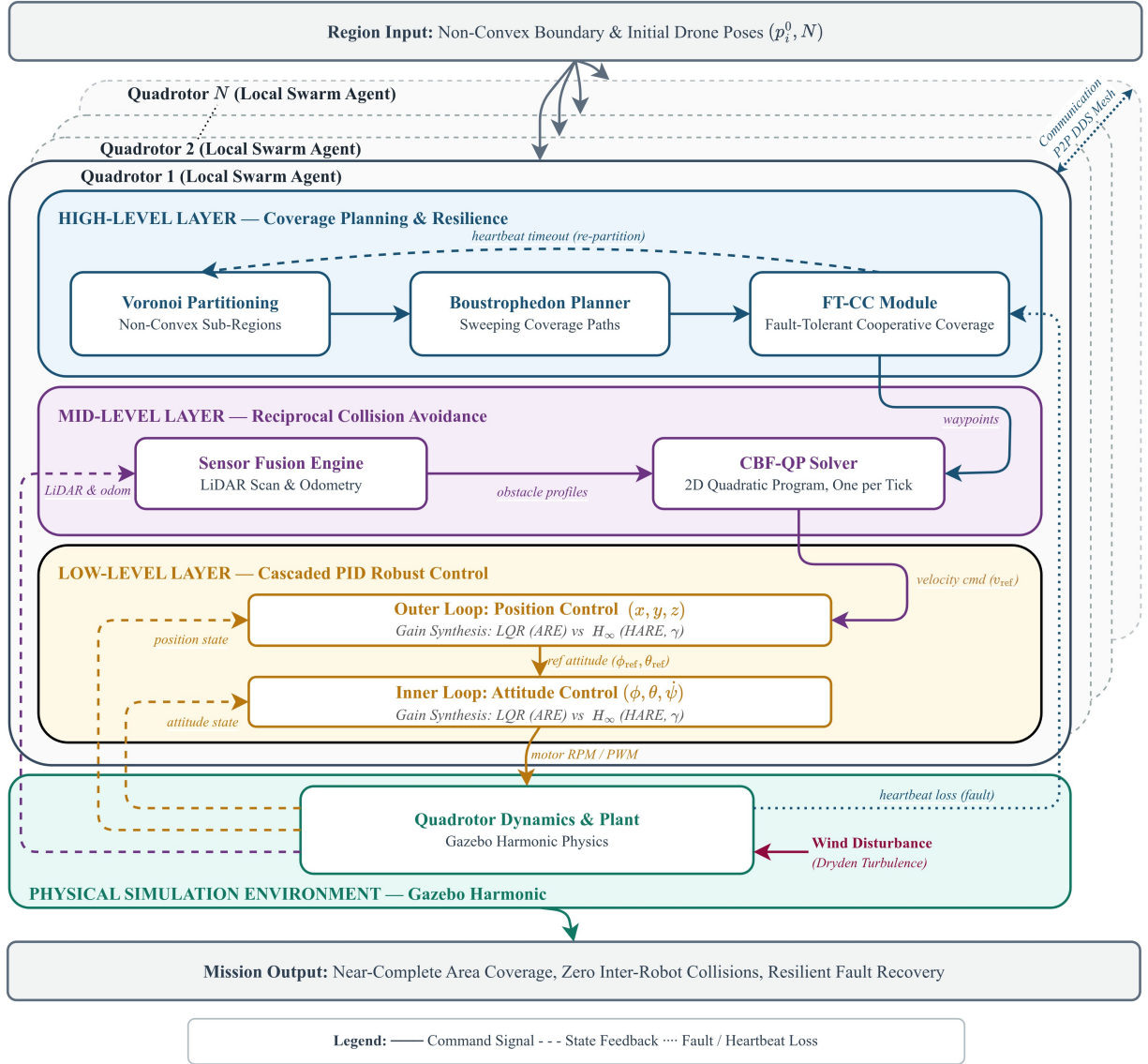


FIGURE 1. Three-layer control architecture for resilient swarm geodetic mapping. The per-agent stacks show how the method is formulated: every vehicle carries the full stack and needs only its neighbours' states, which is what would allow the agents to be distributed. In the study reported here the seven stacks are executed inside one coordinator process, so that the reciprocal responsibility split is computed from a single swarm snapshot. Command signals flow downward, state feedback upward, and heartbeat loss triggers re-partitioning in the FT-CC module.

never writes a velocity command directly: it writes a desired velocity that is filtered through the QP in a single function, so the avoidance layer is separable from the planner.

The stack is formulated per agent. Each vehicle carries its own state machine, its own sweep assignment, and its own two-variable quadratic program, and the constraint rows of that program need nothing beyond the agent's own state, the positions and manoeuvring weights of its neighbours, and the confirmed obstacle set. Nothing in the formulation requires a global optimisation. The present study nevertheless executes all seven agents inside a single coordinator process, because the reciprocal split of Eq. (26) is exact only when both weights are read from the same instant; a deployment that distributes the agents across vehicles would have to work with neighbour weights that arrive over a link and may be stale, which is taken up among the limitations.

High-Level Layer: Coverage Planning and Resilience

Centroidal Voronoi Partitioning

Let $\Omega \subset \mathbb{R}^2$ be the survey polygon, not assumed convex, and $\mathbf{p}_i$ the seed of agent i , $i = 1, \dots, N$. The Voronoi cell of agent i is

$$V_i = \{\mathbf{p} \in \Omega \mid \|\mathbf{p} - \mathbf{p}_i\| \leq \|\mathbf{p} - \mathbf{p}_j\|, \forall j \neq i\}, \quad (1)$$

obtained by successive Sutherland–Hodgman clipping of Ω against the perpendicular bisector H_{ij} of each pair,

$$H_{ij} = \left\{ \mathbf{p} \mid (\mathbf{p}_j - \mathbf{p}_i)^\top \left(\mathbf{p} - \frac{\mathbf{p}_i + \mathbf{p}_j}{2} \right) \leq 0 \right\}, \quad V_i = \Omega \cap \bigcap_{j \neq i} H_{ij}. \quad (2)$$

Cells are equalised by Lloyd iteration [6], moving each seed to the area centroid of its own cell, which is the discrete descent underlying centroidal Voronoi coverage control [5, 7],

$$\mathbf{p}_i^{(k+1)} = \frac{1}{|V_i^{(k)}|} \iint_{V_i^{(k)}} \mathbf{p} \, dA, \quad |V_i^{(k)}| = \iint_{V_i^{(k)}} dA, \quad (3)$$

run for 12 iterations. Clipping acts on the polygon itself and not on a bounding box, so concave boundaries and interior exclusions need no diffeomorphism, geodesic metric, or auxiliary markers [8, 9]. Each cell is then eroded by a margin $\delta_c = 0.35$ m before planning [10].

Sensing Model and Coverage Semantics

The sweep pitch is not free: it is fixed by the mapping payload. For a downward-looking camera of focal length f , pixel pitch p , and sensor width s_w , flown at altitude h_a , the ground sampling distance and the across-track swath are

$$\text{GSD} = \frac{h_a p}{f}, \quad w = \frac{h_a s_w}{f} = 2h_a \tan \frac{\Phi}{2}, \quad (4)$$

with Φ the across-track field of view. Photogrammetric reconstruction requires a side overlap o_s between adjacent strips, which bounds the pitch and hence the effective sensing radius used by the planner,

$$\Delta y \leq w(1 - o_s), \quad r_s = \frac{1}{2}w, \quad (5)$$

so that a cell is fully mapped, not merely overflowed. Coverage in this paper is evaluated against the disc model of Eq. (5): a grid cell counts as observed once its centre falls within r_s of some agent. This is a geometric abstraction of the payload, and it is the same abstraction used by the coverage-control literature [5]. It deliberately does not model radiometric quality, feature matching, or bundle-adjustment residuals, so the reported η quantifies *area coverage at the required GSD*, not final map accuracy; the distinction is revisited among the limitations.

Polygon-Aware Boustrophedon Sweep

Each cell is swept by boustrophedon decomposition [11, 13]. For cell V_i with vertical extent $[y_{\min}, y_{\max}]$ and sweep pitch Δy , the scanline set is $\mathcal{Y}_i = \{y_{\min} + k\Delta y\}$, and each scanline yields the intervals

$$S_i(y) = \{[x_{2k-1}(y), x_{2k}(y)]\}_{k=1}^{m(y)}, \quad \{x_1(y) < \dots < x_{2m}(y)\} = \partial V_i \cap \{\cdot, y\}, \quad (6)$$

where $m(y) > 1$ for a concave cell. A fixed-pitch horizontal sweep leaves an unswept wedge between the last scanline and the boundary at any tapered corner. The planner therefore replaces the extremal rows by the cell's own boundary chains C_i^- (bottom) and C_i^+ (top), keeping interior rows horizontal, giving the flat waypoint list

$$\mathcal{W}_i = C_i^- \cup \bigcup_{y \in \mathcal{Y}_i \setminus \{y_{\min}, y_{\max}\}} \sigma_y(S_i(y)) \cup C_i^+, \quad (7)$$

with σ_y reversing the traversal direction on alternate rows; Figure 2 contrasts the two constructions.

Chain vertices closer than $\ell_{\min} = 0.60$ m are merged: a segment shorter than the braking distance at cruise speed cannot be decelerated into and appears as end-of-row overshoot, while the sensor radius $r_s = 0.95$ m keeps the merged vertices covered. The entry end of C_i^- is selected from the agent's current position.

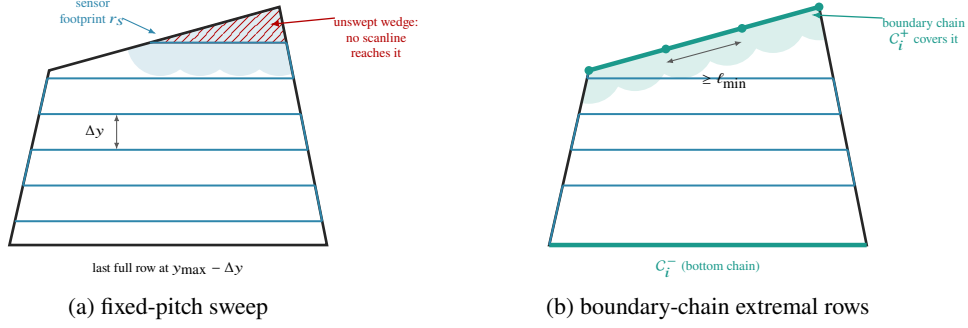


FIGURE 2. Why the extremal sweep rows follow the cell boundary, drawn schematically for a tapered cell. (a) Horizontal scanlines at a fixed pitch Δy stop at the last full row; the sensor footprint of radius r_s (shaded) does not reach the wedge between that row and the boundary, so the wedge is never mapped, and the defect grows with the acuteness of the corner. (b) Replacing the first and last rows by the cell's own boundary chains C_i^- and C_i^+ , Eq. (7), sweeps the boundary itself and closes the wedge by construction, while interior rows stay horizontal. Chain vertices closer than $\ell_{\min}$ are merged, since a segment shorter than the braking distance at cruise speed cannot be decelerated into.

Obstacle Perception and Replanning

Obstacle positions are not supplied to the planner. Each mission starts with an *empty* obstacle map, and the swarm builds it while flying, so the coverage results reported later are those of a system that had to find the cylinders before it could route around them.

Every LiDAR return is first screened: hits that fall near another agent or near the arena wall are discarded, since a neighbouring vehicle subtends an arc indistinguishable from a cylinder. Surviving returns are clustered by angular continuity, and a circle is fitted to each cluster by algebraic least squares, giving a detection $(\mathbf{c}, r)$. Detections from all agents are merged into one shared map. A detection is associated with an existing track when it lies within $d_a = 1.20$ m of it, otherwise it opens a new track. A track becomes *confirmed*, and thus visible to the planner and to the barrier layer, only after $n_{\min} = 4$ sightings,

$$O^{\text{known}}(t) = \{ \tau \mid n_\tau(t) \geq n_{\min} \}, \quad \mathbf{c}_\tau \leftarrow \mathbf{c}_\tau + \frac{1}{n_\tau + 1} (\mathbf{c} - \mathbf{c}_\tau), \quad (8)$$

with the centre and radius kept as running means so that a single noisy frame cannot shift the map. The confirmation threshold is what keeps edge reflections and missed drone rejections from populating the map with phantom cylinders; a track whose centre drifts further than 0.70 m from its anchor is discarded as a moving object and never promoted.

When a track is confirmed, the agents whose swept cells its keep-out disc intersects replan. Only rows *above* the current sweep line are replaced: sweeps run bottom to top, so re-planning a completed row would sweep it twice. Between the moment an obstacle enters the sensing range and the moment its track is confirmed, nothing in the plan knows about it, and safety rests entirely on the barrier layer of the next section, which acts on the same confirmed set O^{known} .

Two obstacle sets must therefore be distinguished, and conflating them would invalidate the results. O^{known} is what the system knows; the planner and every constraint row see only this. The ground-truth table O^{true} is used *only* for scoring — it defines the coverage denominator and detects collisions — because grading an agent against obstacles it has not yet discovered is the entire point of the exercise. No quantity reported in this paper lets O^{true} leak into a control or planning decision.

Obstacle Routing

Routing acts on the confirmed set O^{known} only. For a confirmed obstacle k at $\mathbf{o}_k$ with radius ρ_k and keep-out radius $R^{\text{ko}} = 1.30$ m, every scanline interval is *split* rather than trimmed,

$$[x_a, x_b] \mapsto [x_a, x_b] \setminus \{x \mid \|(x, y) - \mathbf{o}_k\| < R^{\text{ko}}\}, \quad (9)$$

which returns a list, so an obstacle lying mid-row is handled. A final pass over the assembled list inserts circular detour arcs of radius R^{ko} subtending at most 70° per chord, then string-pulls points back out subject to $\|\mathbf{w} - \mathbf{o}_k\| \geq R^{\text{cl}} = 1.10$ m. The chord of a 70° arc dips to $R^{\text{ko}} \cos 35^\circ = 1.065$ m, still above the physical requirement

$\rho_k + r_d + \delta_s = 0.40 + 0.22 + 0.25 = 0.87$ m. Since $R^{\text{ko}} < r_s$, the annulus around each obstacle is mapped during the detour and avoidance costs no coverage. Obstacle k is assigned to every agent whose swept cell its keep-out disc intersects,

$$O_i = \{k \mid \mathcal{B}(\mathbf{o}_k, R^{\text{ko}}) \cap V_i \neq \emptyset\}, \quad (10)$$

not to the single agent whose cell contains it; exclusive assignment leaves a neighbour sweeping through a keep-out zone with no detour planned.

Start-Point Deconfliction

Each agent has two candidate entries $e_i \in \{0, 1\}$, one at each end of C_i^- . Neighbouring cells can place two entries closer than the hard separation d_{hard} , after which the second agent can never arrive. All 2^N assignments are scored lexicographically,

$$\mathbf{e}^* = \arg \max_{\mathbf{e} \in \{0,1\}^N} \min \left(\min_{i < j} \|\mathbf{s}_i(e_i) - \mathbf{s}_j(e_j)\|, d_{\text{cap}} \right), \quad \text{tie-break } \min \sum_{i=1}^N \|\mathbf{p}_i - \mathbf{s}_i(e_i)\|, \quad (11)$$

with $d_{\text{cap}} = 2.5$ m so that separation beyond a comfortable margin is not bought with further transit.

Fault-Tolerant Cooperative Coverage

Agent health is tracked by peer-to-peer heartbeat. Agent k is declared dead when $t - t_k^{\text{hb}} > T_{\text{hb}}$. The map an agent has built is carried on the agent, so losing the vehicle loses the data: on declaring k dead the coordinator *erases* the occupancy record over the whole of V_k rather than crediting the part k had already flown. The recovery region is therefore the union of the complete cells of the dead set,

$$V_{\text{merged}} = \bigcup_{k \in \mathcal{D}} V_k, \quad \mathcal{G}[V_k] \leftarrow \text{unobserved}, \quad \forall k \in \mathcal{D}, \quad (12)$$

with $\mathcal{G}$ the shared occupancy grid. The erasure is deliberately scoped to the dead agent's own cell: area already re-swept by a helper during an earlier recovery is retained, since that data sits on a vehicle still flying. A fresh boustrophedon path is generated over V_{merged} by Eq. (7), and its rows are redistributed to at most three surviving neighbours, which resume their own cells afterwards. Rows still queued on an agent that dies while acting as a helper are recaptured and folded into the next merge, so no row is dropped when a failure lands on a helper. Figure 3 illustrates the sequence.

This is the conservative choice, and it costs real flight time: a failure at high mission progress discards a nearly complete cell. It is also the only sound one, because the coordinator has no way to recover the observations that went down with the vehicle.

Mid-Level Layer: Decentralized Collision Avoidance

A control barrier function certifies forward invariance of a safe set through a single inequality on the control input, and pairing it with a quadratic program yields a minimally invasive safety filter [14, 15]; the reciprocal multi-agent form is due to Wang *et al.* [16]. This section instantiates that construction with a class- $\mathcal{K}$ function that is derived from the vehicle, not tuned.

Closed-Loop Plant Identification

The barrier is built on the identified closed-loop response of the low-level position controller, so no constant in this layer is hand-tuned. With the reference rule $\mathbf{x}_{\text{ref}} = \mathbf{x} + T_{\text{lead}} \mathbf{v}_{\text{cmd}}$ and the outer-loop law of Eq. (36), the commanded-velocity plant is first order with acceleration saturation,

$$\dot{v} = \text{sat}_{\pm a_{\text{max}}} (k_v (v_{\text{cmd}} - v)), \quad (13)$$

$$k_v = g K_d, \quad a_{\text{max}} = g \tan \theta_{\text{max}}, \quad T_{\text{lead}} = \frac{K_d - k_{\text{ff}}}{K_p}, \quad v_c = \frac{a_{\text{max}}}{k_v}, \quad (14)$$

where T_{lead} is chosen so that the DC gain from v_{cmd} to v is unity and v_c is the speed above which the saturation binds. Integrating Eq. (13) gives the exact stopping distance from speed s ,

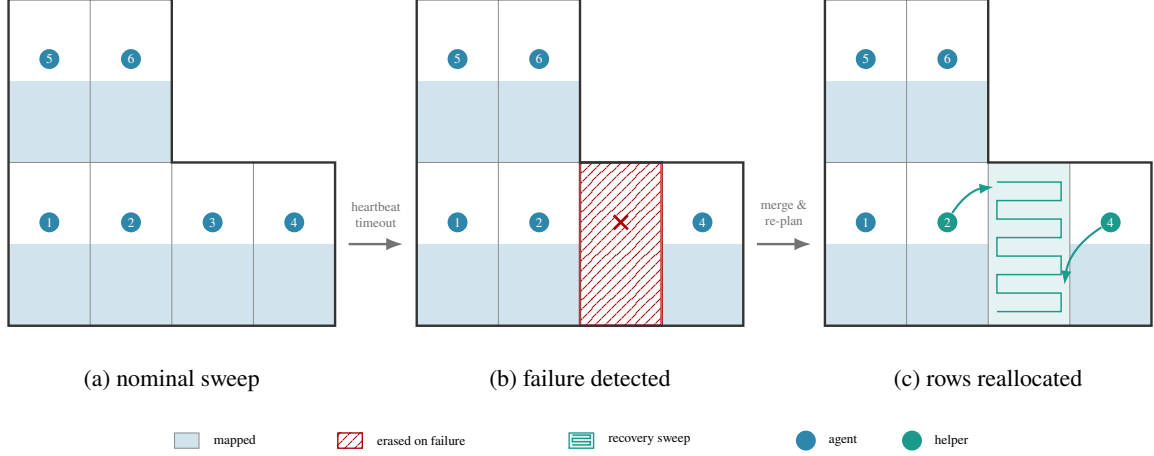


FIGURE 3. Fault-tolerant cooperative coverage, drawn schematically. (a) Each agent sweeps its own cell and marks the shared occupancy grid as it goes. (b) Agent 3 stops sending heartbeats; because the map it had accumulated was carried on that vehicle, the coordinator erases the occupancy record over the whole of V_3 rather than crediting the part already flown, so the entire cell reverts to unobserved. (c) A fresh boustrophedon path is generated over V_{merged} , Eq. (12), and its rows are split between a bounded set of surviving neighbours, which resume their own cells once the recovery rows are finished.

$$D(s) = \begin{cases} \frac{s^2 + v_c^2}{2a_{\max}}, & s > v_c, \\ \frac{s}{k_v}, & s \leq v_c. \end{cases} \quad (15)$$

Actuator-Consistent Class- $\mathcal{K}$ Function

Let h be the clearance to a constraint surface and T_d the control dead time. The admissible approach speed is the largest s that can still be brought to rest within h ,

$$s T_d + D(s) \leq h \iff s^2 + 2a T_d s + (v_c^2 - 2ah) \leq 0, \quad (16)$$

whose positive root gives the class- $\mathcal{K}$ function

$$\varphi(h) = \max\left(0, -a T_d + \sqrt{\max(0, a^2 T_d^2 + 2ah - v_c^2)}\right). \quad (17)$$

Its inverse, the reaction distance required at closing speed s , is

$$\varphi^{-1}(s) = \frac{s^2 + 2a T_d s + v_c^2}{2a}, \quad h_0 = \frac{v_c^2 - a^2 T_d^2}{2a}, \quad (18)$$

with h_0 the largest clearance at which $\varphi = 0$ and the vehicle must already be stopped. Two properties are used downstream: φ is monotone increasing and vanishes for $h \leq h_0$, hence a valid class- $\mathcal{K}$ function; and it is Lipschitz with constant at most $1/T_d$, so the constraint set never tightens faster than a per second, giving recursive feasibility with respect to the rate limits of Eq. (23). A linear $\alpha(h) = \gamma h$ cannot reproduce the flat zero region below h_0 and is therefore inadmissible near $h \rightarrow 0$. Figure 4 shows both properties.

The acceleration budget per constraint class c is

$$a_{\text{eff}}^{(c)} = \max\left(0.15 a_{\max}, \mu_c a_{\max} - |a_{\text{obs}}| - |a_{\text{wind}}|\right), \quad (19)$$

reserving margin for obstacle acceleration and wind so the row stays satisfiable inside the rate box.

Constraint Rows

Every constraint has the form $h = d - R$ with derivative $\dot{h} = \hat{\mathbf{n}}^T(\mathbf{u} - \mathbf{v}_{\text{obs}})$ and $\hat{\mathbf{n}} = (\mathbf{p} - \mathbf{o})/\|\mathbf{p} - \mathbf{o}\|$. The CBF

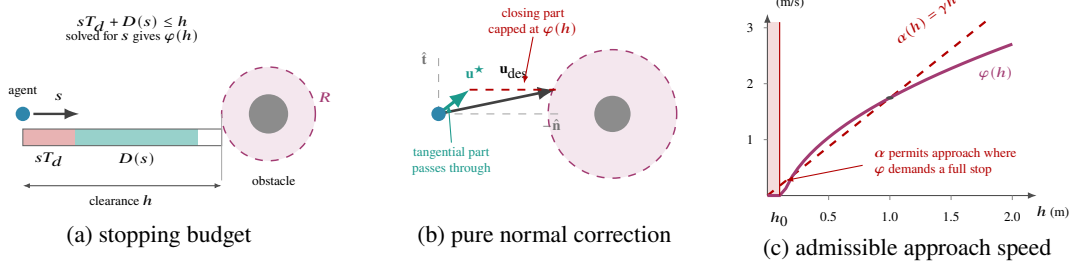


FIGURE 4. The barrier layer, drawn schematically. (a) The admissible approach speed follows from a distance budget: travel during the dead time plus the exact stopping distance of Eq. (17) must fit inside the clearance, and solving that inequality for s yields $\varphi(h)$. (b) Because the constraint row normal is $-\hat{\mathbf{n}}$, the quadratic program caps only the component of the commanded velocity that closes on the obstacle; the tangential component passes through unchanged, so the agent skirts the obstacle while still making progress. (c) $\varphi(h)$ evaluated at $a = 2.5 \text{ m/s}^2$, $T_d = 0.15 \text{ s}$, $v_c = 0.80 \text{ m/s}$. Below h_0 the admissible speed is exactly zero, a flat region no linear $\alpha(h) = \gamma h$ can reproduce: matching the two at one clearance (marked) necessarily over-permits nearer in.

condition $\dot{h} \geq -\varphi(h)$ becomes a linear row in the commanded velocity $\mathbf{u}$. For obstacle k ,

$$-\hat{\mathbf{n}}_k^\top \mathbf{u} \leq \varphi(h_k, a_{\text{eff}}^{(c)}, T_d, v_c) - \hat{\mathbf{n}}_k^\top \mathbf{v}_k, \quad h_k = \|\mathbf{p} - \mathbf{o}_k\| - (\rho_k + r_d + \delta_c). \quad (20)$$

Because the row normal is $-\hat{\mathbf{n}}_k$, the active-constraint solution is a pure normal correction, $\mathbf{u}^* = \mathbf{v}_{\text{nom}} + (\beta - \hat{\mathbf{n}}_k^\top \mathbf{v}_{\text{nom}}) \hat{\mathbf{n}}_k$: the tangential component passes through unchanged and the agent skirts the obstacle while still making progress. Inter-agent rows are reciprocal, with responsibility split $\lambda_{ij} + \lambda_{ji} = 1$ weighted by mission state as set out below, and are issued at two radii so the soft tier can be released when cells legitimately place agents close together,

$$-\hat{\mathbf{n}}_{ij}^\top \mathbf{u}_i \leq \lambda_{ij} \varphi(d_{ij} - R_\bullet), \quad R_\bullet \in \{R_{\text{hard}}, R_{\text{soft}}\}. \quad (21)$$

Arena walls and sweep-row stop planes use the outward normal directly,

$$\hat{\mathbf{a}}_w^\top \mathbf{u} \leq \varphi(h_w), \quad h_w = \text{signed distance to the plane}, \quad (22)$$

which replaces the hand-tuned end-of-row feedforward ramp: zero overshoot follows from the same theorem, not from a tuned constant. Speed and rate limits are inscribed regular n -gons, guaranteed never to overstate the true disc,

$$A_\Pi(\mathbf{c}, r) = [\cos \vartheta_l, \sin \vartheta_l]_{l=1}^n, \quad b_{\Pi, l} = r \cos \frac{\pi}{n} + A_{\Pi, l} \mathbf{c}, \quad \vartheta_l = \frac{2\pi l}{n}, \quad (23)$$

evaluated as $\Pi(\mathbf{0}, v_{\text{max}})$ for speed and $\Pi(\mathbf{u}_{\text{prev}}, a_{\text{max}} \Delta t)$ for the rate box.

Quadratic Program and Solver

Stacking all rows into $\mathbf{A}\mathbf{u} \leq \mathbf{b}$, each agent solves once per tick

$$\mathbf{u}^* = \arg \min_{\mathbf{u} \in \mathbb{R}^2} \frac{1}{2} \|\mathbf{u} - \mathbf{u}_{\text{des}}\|^2 \quad \text{s.t.} \quad \mathbf{A}\mathbf{u} \leq \mathbf{b}, \quad (24)$$

a Euclidean projection onto a polyhedron in $\mathbb{R}^2$. In two dimensions the optimum lies at $\mathbf{u}_{\text{des}}$ when feasible, on a single constraint line, or at the intersection of a pair, so the complete KKT candidate set

$$\mathcal{U} = \{\mathbf{u}_{\text{des}}\} \cup \left\{ \mathbf{u}_{\text{des}} - \frac{A_l \mathbf{u}_{\text{des}} - b_l}{\|A_l\|^2} A_l^\top \right\}_{l=1}^m \cup \left\{ \mathbf{u} : A_l \mathbf{u} = b_l, A_{l'} \mathbf{u} = b_{l'} \right\}_{l < l'}, \quad (25)$$

of $1 + m + \binom{m}{2}$ points is enumerated, filtered for feasibility, and minimised over. The result is exact rather than iterative, with no convergence tolerance and no silent failure. In this study all agents are solved from a single swarm snapshot, which is what makes the reciprocal split exact; the rows themselves are per agent, so the same formulation admits a distributed implementation once stale neighbour weights are accounted for.

Each agent carries a manoeuvring willingness w_i fixed by its mission state, and a pair splits the avoidance burden in proportion to it,

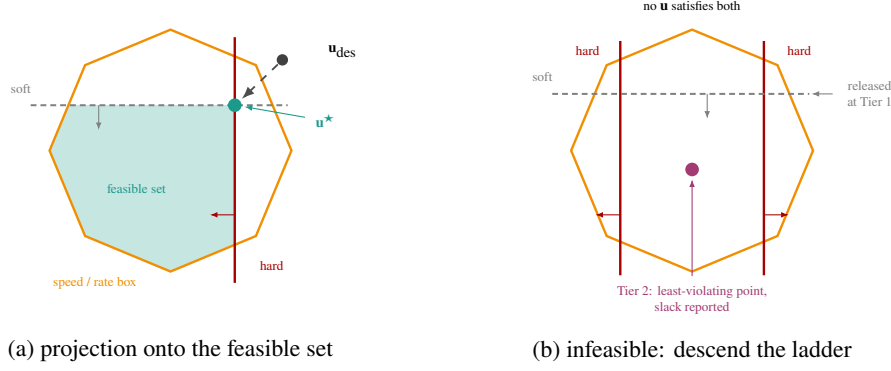


FIGURE 5. Geometry of the safety filter in velocity space, drawn schematically. (a) Each obstacle, neighbour, wall and stop plane contributes one half-plane, and the speed and rate limits contribute the inscribed octagon of Eq. (23); arrows mark the admissible side. The command returned is the Euclidean projection of the planner’s desired velocity onto their intersection, Eq. (24), computed exactly by enumerating the candidate vertices of Eq. (25) — here the projection lands on the corner where two rows are simultaneously active. (b) Two opposing hard rows can leave the intersection empty; the ladder then releases soft rows in a fixed order and, failing that, returns the least-violating point inside the kinematic box with its residual reported as slack. Hard rows and the kinematic box are never released, so no tier can command a velocity the vehicle cannot produce.

$$\lambda_{ij} = \frac{w_i}{w_i + w_j}, \quad \text{giving} \quad \lambda_{ij} + \lambda_{ji} = 1 \quad \text{exactly}, \quad (26)$$

so summing the two agents’ rows recovers the joint condition $\dot{h} \geq -\varphi(h)$ with no communication or negotiation. The weights encode who gives way. An agent sweeping a row or holding its start yaw takes $w = 0.25$, so it is protected and its mapped line stays straight. An agent pausing at a row end takes $w = 0.50$, and one in transit, stepping between rows, or returning to its centroid takes $w = 1.00$. An agent that has finished, or is waiting for the swarm, takes $w = 1.50$ and yields to everyone.

Feasibility is not assumed. Constraint classes are partitioned in advance into *hard* (static obstacles, moving obstacles, the minimum inter-agent radius), *soft* (arena walls, the comfortable inter-agent radius, sweep-row stop planes), and *kinematic* (the speed and rate boxes), and the solver descends a fixed ladder:

- Tier 0.** Every row is satisfied and the projection of Eq. (24) is returned unchanged. This is the normal operating point.
- Tier 1.** Soft classes are released one at a time in the fixed order walls, comfortable radius, stop planes, stopping at the first release that restores feasibility. Hard classes are never released.
- Tier 2.** Still infeasible: return the command minimising the worst hard row violation while staying inside the kinematic box, $\min_{\mathbf{u}} \max_l (A_l \mathbf{u} - b_l)$ over the hard rows subject to the speed and rate rows. The residual is reported as slack, not hidden.
- Tier 3.** Last resort: decay the previous command, $\mathbf{u} = 0.9 \mathbf{u}_{\text{prev}}$, so an unbounded or undefined velocity is never returned.

The speed and rate boxes are never dropped at any tier, so no tier can command a velocity the vehicle cannot produce. Figure 5 shows both cases.

The fraction of ticks resolved above Tier 0 is reported as $P(\text{tier} > 0)$; a value near zero means the ladder is a safety net, not an operating mode. Deadlock recovery adds a locked tangential bias to the objective target only,

$$\mathbf{u}_{\text{des}} \leftarrow \mathbf{u}_{\text{des}} + \kappa \varsigma_{ik} \hat{\mathbf{t}}_{ik}, \quad \hat{\mathbf{t}}_{ik} = [-\hat{n}_{ik,y}, \hat{n}_{ik,x}]^T, \quad \varsigma_{ik} \in \{-1, +1\}, \quad (27)$$

with the side ς_{ik} latched per (agent, obstacle) pair to prevent chattering. Neither A nor $\mathbf{b}$ is ever modified, so liveness is never traded against safety.

Low-Level Layer: Cascaded PID Robust Control

Model and Decomposition

With mass m , inertia $\text{diag}(I_x, I_y, I_z)$, thrust T , and body torques (τ_x, τ_y, τ_z) , the rigid-body dynamics are [18]

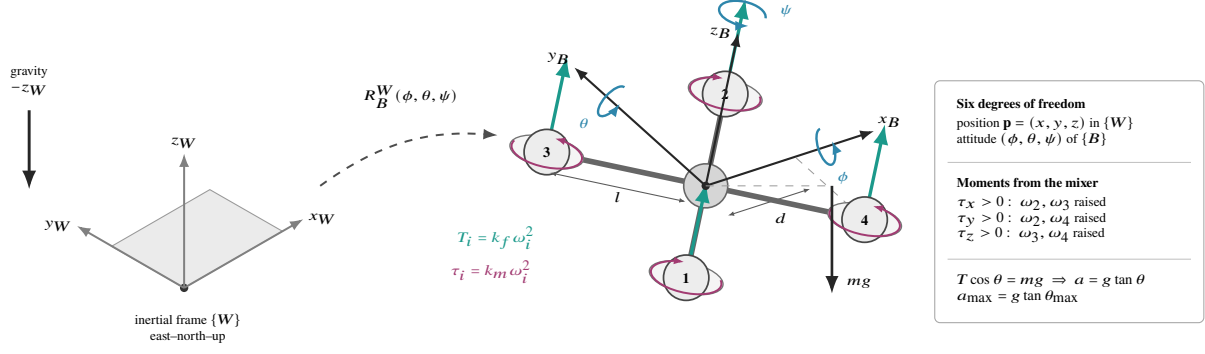


FIGURE 6. Free-body diagram of the quadrotor and the conventions used throughout. The inertial frame $\{W\}$ is east–north–up, so gravity acts along $-z_W$ and the translational dynamics take the form of Eq. (28); $R_B^W(\phi, \theta, \psi)$ maps body to world; the weight acts at the centre of mass, marked by the dot. Roll ϕ is about x_B , pitch θ about y_B and yaw ψ about z_B shown in their positive senses. Rotor i produces thrust $T_i = k_f \omega_i^2$ along $+z_B$ and a reaction torque $\tau_i = k_m \omega_i^2$ opposing its own rotation; the diagonal pairs $\{1, 2\}$ and $\{3, 4\}$ counter-rotate, so the reaction torques cancel in hover and net yaw comes from unbalancing them. The dashed construction resolves the arm l into its components $d = l \sin 45^\circ$ along x_B and $-y_B$, the moment arm over which differential thrust produces roll and pitch. Because altitude is held, $T \cos \theta = mg$ and the tilt limit caps lateral acceleration at $a_{\max} = g \tan \theta_{\max}$ — the bound on which the barrier of Eq. (17) is built, which is what ties the safety layer to the vehicle rather than to a tuning constant. Aerodynamic drag is not shown because it is not modelled here; its omission is revisited among the limitations.

$$m\ddot{\mathbf{p}} = \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} + R(\phi, \theta, \psi) \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix}, \quad I\dot{\omega} + \omega \times I\omega = \tau. \quad (28)$$

Figure 6 is the free-body diagram and fixes the conventions used throughout.

Linearising about hover ($\phi, \theta \approx 0$, $T \approx mg$) decouples the system into four double-integrator subsystems, each of the form $\dot{\mathbf{x}} = A\mathbf{x} + Bu$, $y = C\mathbf{x}$ with

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad B \in \left\{ \begin{bmatrix} 0 \\ g \end{bmatrix}, \begin{bmatrix} 0 \\ -g \end{bmatrix}, \begin{bmatrix} 0 \\ 1/I_y \end{bmatrix}, \begin{bmatrix} 0 \\ 1/I_x \end{bmatrix}, \begin{bmatrix} 0 \\ 1/m \end{bmatrix}, \begin{bmatrix} 0 \\ 1/I_z \end{bmatrix} \right\}, \quad (29)$$

for the longitudinal outer ($x \rightarrow \theta_{\text{ref}}$), lateral outer ($y \rightarrow \phi_{\text{ref}}$), pitch inner ($\theta \rightarrow \tau_y$), roll inner ($\phi \rightarrow \tau_x$), altitude ($z \rightarrow T$), and yaw ($\psi \rightarrow \tau_z$) loops respectively. Design weights are listed in Table 1.

Gain Synthesis

Each loop is augmented with an integrator state,

$$A_a = \begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}, \quad B_a = \begin{bmatrix} 0 \\ I \end{bmatrix}, \quad Q_a = \text{blkdiag}(Q, 0.1 I), \quad R_a = R I. \quad (30)$$

PID-LQR solves the continuous algebraic Riccati equation (ARE),

$$A_a^\top P + P A_a + Q_a - P B_a R_a^{-1} B_a^\top P = 0, \quad K_a = R_a^{-1} B_a^\top P. \quad (31)$$

PID- H_∞ adds an explicit disturbance channel B_w and solves the H_∞ Riccati equation (HARE) [19] instead,

$$A_a^\top P + P A_a + Q_a - P \tilde{B} \tilde{R}^{-1} \tilde{B}^\top P = 0, \quad \tilde{B} = \begin{bmatrix} B_a & B_w \end{bmatrix}, \quad \tilde{R} = \begin{bmatrix} R_a & 0 \\ 0 & -\gamma^2 I \end{bmatrix}, \quad (32)$$

so that the closed loop satisfies $\|T_{zw}\|_\infty < \gamma$. Both syntheses produce a state-feedback gain K_a . It is mapped onto the PID triple through the structure matrix

$$\Gamma = \begin{bmatrix} C^\top & (CA)^\top & (CA^2)^\top \\ 0 & (CB)^\top & (CAB)^\top \end{bmatrix}, \quad \hat{K} = \Gamma^\dagger K_a^\top = \begin{bmatrix} \hat{k}_1 & \hat{k}_2 & \hat{k}_3 \end{bmatrix}^\top, \quad (33)$$

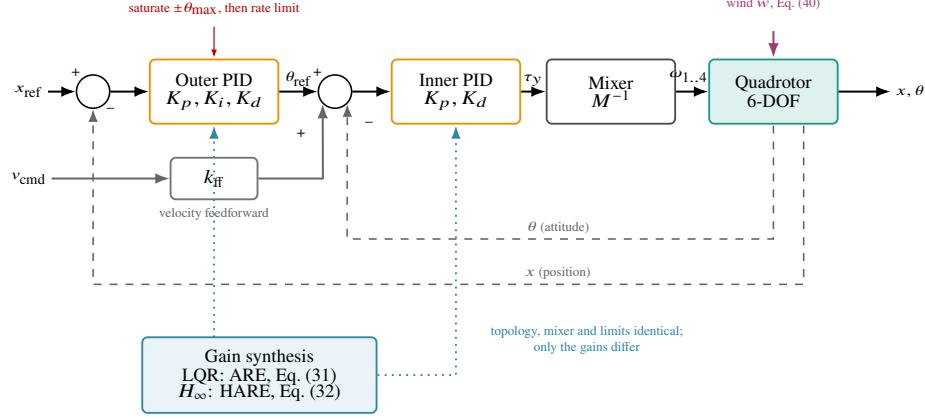


FIGURE 7. Low-level cascade, shown for the longitudinal axis; the lateral, altitude, and yaw subsystems have the same structure with the input matrices of Eq. (29). The outer loop maps position error to a reference attitude, which is saturated at the tilt limit and rate limited before the inner loop converts attitude error to torque; the mixer inverts Eq. (39) to rotor speeds. Both controllers under comparison share this topology, mixer, and limits exactly and differ only in the shaded synthesis block, so any difference in tracking is attributable to gain synthesis alone.

using the Moore–Penrose pseudoinverse, followed by the back-substitution that removes the algebraic loop introduced by the derivative term,

$$K_d = \hat{k}_3(I + CB\hat{k}_3)^{-1}, \quad K_i = (I - K_dCB)\hat{k}_1, \quad K_p = (I - K_dCB)\hat{k}_2. \quad (34)$$

Both controllers share the identical cascade topology, mixer, and limits, differing only in whether Eq. (31) or Eq. (32) is solved; any difference in tracking is therefore attributable to gain synthesis alone.

Selecting the Attenuation Bound

In Eq. (32) the disturbance enters only through $-\gamma^{-2}B_wB_w^\top$. As γ grows this term shrinks quadratically until it is negligible beside $B_aR_a^{-1}B_a^\top$, at which point HARE degenerates into ARE and the H_∞ controller returns gains numerically indistinguishable from LQR. The bound is therefore placed near its feasibility limit: $\gamma_{\min}$ is located per subsystem by bisection on the existence of a stabilising solution to Eq. (32), and the operating value is set to

$$\gamma = 1.3 \gamma_{\min}. \quad (35)$$

Table 2 lists both. An earlier iteration of this work used $\gamma_{\text{out}} = 80$, $\gamma_{\text{in}} = 64$, which made the disturbance term 10^3 to 5×10^4 times smaller than the control term; the two controllers then agreed to within 2% on every gain and produced near-identical trajectories across a full mission set.

Control Law and Mixing

Figure 7 shows the resulting cascade.

With body-frame position errors e_x^b, e_y^b and measured body velocities v_x^b, v_y^b , the outer loops generate reference attitudes with velocity feedforward $k_{\text{ff}} = 0.15$ rad per m/s,

$$\theta_{\text{ref}} = \text{sat}_{\pm\theta_{\max}}\left(K_p^x e_x^b + K_i^x \int e_x^b - K_d^x v_x^b + k_{\text{ff}} v_{x,\text{cmd}}^b\right), \quad \phi_{\text{ref}} = \text{sat}_{\pm\theta_{\max}}\left(K_p^y e_y^b + K_i^y \int e_y^b - K_d^y v_y^b - k_{\text{ff}} v_{y,\text{cmd}}^b\right), \quad (36)$$

both rate-limited before use. The inner loops, altitude loop, and yaw loop give

$$\tau_y = K_p^\theta(\theta_{\text{ref}} - \theta) - K_d^\theta \dot{\theta}, \quad \tau_x = K_p^\phi(\phi_{\text{ref}} - \phi) - K_d^\phi \dot{\phi}, \quad \tau_z = K_p^\psi e_\psi - K_d^\psi \dot{\psi} + k_{\text{ff}}^\psi \dot{\psi}_{\text{cmd}}, \quad (37)$$

$$T = \frac{K_p^z e_z + K_i^z \int e_z + K_d^z \dot{e}_z + mg}{\cos \phi \cos \theta}, \quad (38)$$

TABLE 1. Design weights per subsystem, Eqs. (29)–(30).

Subsystem / loop	B	Q	R
Longitudinal x , outer	$[0, g]^\top$	$\text{diag}(0.05, 0.30)$	1.00
Pitch θ , inner	$[0, 1/I_y]^\top$	$\text{diag}(2.0, 0.2)$	0.08
Lateral y , outer	$[0, -g]^\top$	$\text{diag}(0.05, 0.30)$	1.00
Roll ϕ , inner	$[0, 1/I_x]^\top$	$\text{diag}(2.0, 0.2)$	0.08
Altitude z	$[0, 1/m]^\top$	$\text{diag}(1, 1000)$	1.00
Yaw ψ	$[0, 1/I_z]^\top$	$\text{diag}(1.0, 4.0)$	4.00

TABLE 2. H_∞ attenuation bounds. $\gamma_{\min}$ is the feasibility limit of Eq. (32); the operating value is Eq. (35).

Subsystem	$\gamma_{\min}$	γ
Outer position x, y	2.54	3.30
Inner attitude ϕ, θ	2.54	3.31
Altitude z	4.61	6.00
Yaw ψ	9.02	11.73

where the denominator is the tilt compensation that keeps vertical thrust constant during manoeuvres. Rotor speeds follow by inverting the mixer, with $d = l \sin 45^\circ$,

$$\begin{bmatrix} T \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \underbrace{\begin{bmatrix} k_f & k_f & k_f & k_f \\ -k_f d & k_f d & k_f d & -k_f d \\ -k_f d & k_f d & -k_f d & k_f d \\ -k_m & -k_m & k_m & k_m \end{bmatrix}}_M \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix}, \quad \omega = \sqrt{\text{sat}_{[0, \omega_{\max}^2]}(M^{-1}\mathbf{U})}. \quad (39)$$

Disturbance Model

Scheme 2 injects a first-order Gauss–Markov gust model approximating Dryden turbulence [18, 20]. The full Dryden model is a shaping filter whose spectrum is parameterised by altitude-dependent scale lengths and airspeed; at the low altitude and low airspeed of a mapping sweep the along-axis component is well represented by its first-order approximation,

$$w_{k+1} = \alpha_w w_k + \beta_w \varepsilon_k, \quad \alpha_w = e^{-\Delta t / \tau_w}, \quad \beta_w = \sigma_w \sqrt{1 - \alpha_w^2}, \quad \varepsilon_k \sim \mathcal{N}(0, 1), \quad (40)$$

with $\sigma_w = 2.5$ and $\tau_w = 0.5$ s, applied independently per axis. This is the w channel that B_w in Eq. (32) represents.

Simulation Setup

Experiments run in ROS 2 with the Gazebo physics engine, simulating seven quadrotors with full six-degree-of-freedom rigid-body dynamics, collision geometry, rotor thrust and torque models, and a first-order motor lag. Parameters are listed in Table 3. The airframe values are those of the 3DR Iris as distributed with PX4, but the PX4 firmware itself is not in the loop: there is no EKF2 estimator, commander, mixer, or MAVLink transport, and the controllers under test read vehicle odometry and command rotor speeds directly through Eq. (39). This isolates the gain-synthesis comparison, since the PX4 cascade would otherwise dominate the closed-loop response.

Scenarios: A Load Ladder

The evaluation adds one stressor at a time, so that any change in behaviour is attributable to the stressor just introduced, not to an untangleable combination. Four conditions are flown, each over the same three non-convex regions — L-shape, U-shape, plus-shape — and with both controllers:

1. *Nominal*: no disturbance, no obstacles. Establishes the coverage capability of the planner on non-convex geometry and provides the reference against which every Δ below is measured.
2. *Turbulence*: Gauss–Markov gusts, Eq. (40), no obstacles. Isolates the disturbance-rejection contrast between LQR and H_∞ .
3. *Turbulence and obstacles*: as above, plus nine static cylinders placed inside each region, which activate the planner-side routing of Eqs. (9)–(10). Moving obstacles are excluded; see the limitations.
4. *Turbulence, obstacles, and agent loss*: condition 3 with two agents terminated sequentially, the first once coverage reaches $\eta_1 = 18\%$ and the second at $\eta_2 = 32\%$. The second victim is chosen among the helpers already reassigned by the first, so that the compound-polygon merge of Eq. (12) and the orphan-row rescue are both exercised rather than only a single clean reassignment.

Conditions 3 and 4 differ by the failures alone, which is what makes the coverage loss attributable to agent loss, not to the obstacle field. Every mission logs per-agent telemetry to CSV; all reported values derive from those logs.

TABLE 3. Airframe and actuator parameters.

Quantity	Sym.	Value
Mass	m	1.50 kg
Roll/pitch inertia	$I_x = I_y$	0.0291 kg m ²
Yaw inertia	I_z	0.0552 kg m ²
Arm length	l	0.22 m
Maximum tilt	$\theta_{\max}$	15 deg
Thrust coefficient	k_f	8.549e-6 N/(rad/s) ²
Torque coefficient	k_m	1.368e-7 N m/(rad/s) ²
Motor time constant	τ_m	0.02 s
Max. rotor speed	$\omega_{\max}$	1100 rad/s
Roll/pitch torque cap	$\tau_{rp}^{\max}$	0.80 N m
Yaw torque cap	$\tau_y^{\max}$	0.08 N m

TABLE 4. Control, planner, and mission parameters.

Quantity	Sym.	Value
Attitude feedforward	k_{ff}	0.15 rad/(m/s)
Reference lead time	T_{lead}	0.3287 s
Control dead time	T_d	0.15 s
Sweep speed	v_{sweep}	1.6 m/s
Command speed cap	$v_{\max}$	3.0 m/s
Sensor radius	r_s	0.95 m
Obstacle keep-out	R^{ko}	1.30 m
Voronoi clip margin	δ_c	0.35 m
V2V hard / soft	$R_{h/s}$	0.70/1.20 m
Track confirmation	$n_{\min}$	4 hits
Wind σ_w, τ_w	—	2.5, 0.5 N, s
Number of agents	N	7

Repetitions and Statistical Treatment

Only the gust sequence is stochastic, and the seed enters nothing else, so the planned paths are identical across repetitions of a given cell. Conditions 2 and 3 are therefore flown $N_r = 5$ times per region and controller, and condition 4 $N_r = 3$ times; condition 1 is deterministic given the initial poses and is flown once, its repetitions serving only to confirm simulator reproducibility. Results are reported as mean $\pm$ sample standard deviation. The LQR versus H_∞ contrast is tested on the paired per-(region, seed) difference with a two-sided Wilcoxon signed-rank test, paired because both controllers see the identical gust sequence for a given seed. This gives $6 + 30 + 30 + 18 = 84$ missions, of which the 12 for conditions 1 and 2 are already available.

[TODO: N_r di atas adalah spesifikasi rancangan. Data yang ADA sekarang baru kondisi 1 dan 2 pada $N_r = 1$; kondisi 3 dan 4 belum dijalankan sama sekali. Sebelum submit: jalankan 48 misi sisanya, lalu isi seluruh tabel dengan mean $\pm$ sd dan nilai p . Kalau waktunya tidak cukup, TURUNKAN klaim N_r di sini dan hapus kolom sd serta uji statistiknya — jangan pernah melaporkan sd dari satu run.]

Metrics

Coverage is the fraction of the region occupancy grid observed by at least one agent,

$$\eta = \frac{1}{|\mathcal{G}|} \left| \left\{ \mathbf{c} \in \mathcal{G} \mid \exists i, t : \|\mathbf{c} - \mathbf{p}_i(t)\| \leq r_s \right\} \right|. \quad (41)$$

For a sweep row through $\mathbf{q}$ with unit direction $\hat{\mathbf{u}}$, the cross-track and longitudinal lag components of the tracking error are

$$e_{\text{ct}} = \|(\mathbf{p} - \mathbf{q}) - ((\mathbf{p} - \mathbf{q})^\top \hat{\mathbf{u}}) \hat{\mathbf{u}}\|, \quad e_{\text{lon}} = (\mathbf{p}_{\text{ref}} - \mathbf{p})^\top \hat{\mathbf{u}}, \quad (42)$$

the former determining map quality and the latter reflecting the reference-lead policy of Eq. (14). Safety and effort are summarised by

$$d_{\min} = \min_t \min_{i < j} \|\mathbf{p}_i(t) - \mathbf{p}_j(t)\|, \quad J_\tau = \int_0^{T_f} \|\boldsymbol{\tau}(t)\|_1 dt, \quad P_{\text{tier} > 0} = \frac{1}{N_k} \sum_k \mathbb{1}[\text{tier}_k > 0]. \quad (43)$$

For the fault-injection scenario, let t_f be the instant an agent is terminated, t_d the instant the coordinator declares it dead on heartbeat timeout, and t_r the instant the first helper begins sweeping a recovery row. The recovery metrics are the detection latency, the recovery start delay, and the coverage loss against the nominal mission over the same region,

$$\Delta t_{\text{det}} = t_d - t_f, \quad \Delta t_{\text{rec}} = t_r - t_f, \quad \Delta \eta = \eta_{\text{nom}} - \eta_{\text{fault}}, \quad (44)$$

with $\Delta \eta$ expressed in percentage points.

Attitude metrics are computed only over the sweeping phase: samples where the yaw reference has been stable for at least 1.5 s and the vehicle exceeds 0.6 m/s. Whole-mission percentiles are dominated by the 180° pivots at row ends and would misattribute a normal turn to a disturbance-induced excursion.

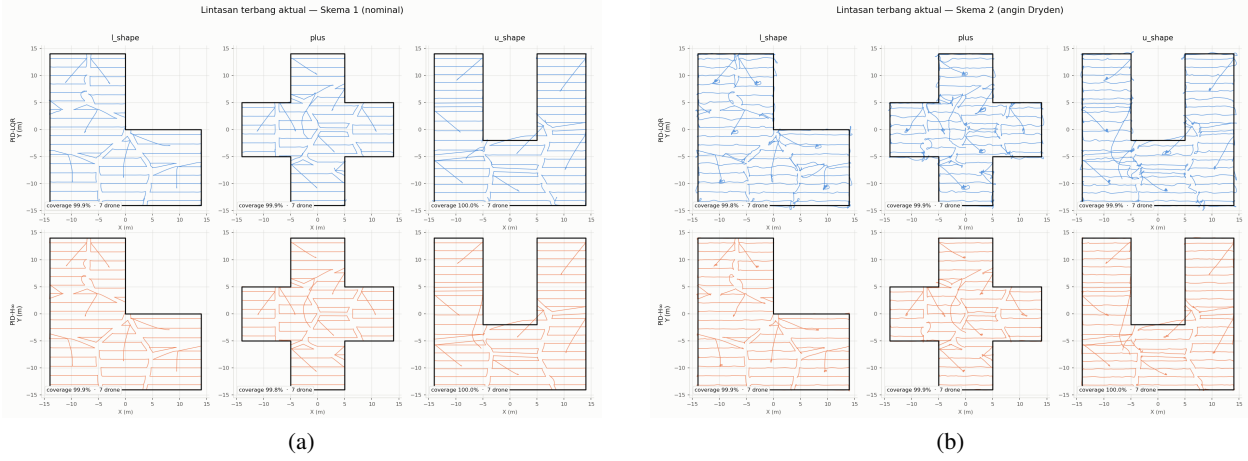


FIGURE 8. Flown trajectories for each non-convex region under (a) the nominal condition and (b) turbulence. Cell boundaries from the centroidal Voronoi partition are overlaid; the first and last sweep row of each cell follows the cell boundary rather than a horizontal line.

TABLE 5. Nominal coverage, flown.

Region	η (%)	Dur. (s)	Path (m)	$d_{\min}$ (m)
L-shape	--	--	--	--
U-shape	--	--	--	--
Plus	--	--	--	--

TABLE 6. Planner-level ablation: geometric coverage of the planned path, computed without simulation.

Region	Fixed pitch	Bnd. chain
L-shape	--	--
U-shape	--	--
Plus	--	--

RESULTS AND DISCUSSION

Results follow the load ladder of the previous section. Each subsection adds one stressor and reports what degrades, what holds, and at what cost.

Coverage Performance on Non-Convex Regions

Table 5 reports the nominal condition and Fig. 8(a) the flown tracks. **[TODO: isi kondisi 1 dari sweep s2_gamma (pasca perbaikan gamma)]**

The coverage figure alone does not establish that the boundary-chain rows are what closes the tapered corners, since a sweep that simply missed them would still report a high number. The right-hand columns therefore compare, at planner level and without flying, the coverage of the path generated by Eq. (7) against a conventional fixed-pitch sweep over the identical cells. **[TODO: jalankan ablasi perencanaan (offline, tanpa Gazebo) dan isi Tabel 6; inilah satu-satunya bukti langsung untuk kontribusi (i)]**

Tracking Performance under Turbulence

Coverage is expected to survive turbulence because the sweep reference is pinned to the row line; what degrades is tracking. Table 7 reports the contrast. **[TODO: isi dari kondisi 2]**

Whatever separation appears here follows from the attenuation bound of Eq. (35). Table 8 makes that dependence concrete: with γ set far above $\gamma_{\min}$, the synthesised H_{∞} gains collapse onto the LQR gains, and the comparison measures nothing. These are properties of the synthesis, obtained without simulation.

[TODO: isi Tabel 8 dengan menjalankan solver pada kedua nilai γ dan membandingkan gain-nya terhadap LQR — murni numerik, tanpa Gazebo.]

Coverage and Safety with Static Obstacles

Nine static cylinders per region activate the planner-side routing and load the barrier layer while the wind is still acting. Table 10 reports the cost of the detours and the safety margins that survive them, and Fig. 11 the resulting tracks. **[TODO: isi dari kondisi 3 (30 misi)]**

TABLE 7. Tracking accuracy and control effort under turbulence, aggregated over the three regions; mean $\pm$ sd over $N_r = 5$ seeds. p is the two-sided Wilcoxon signed-rank test on the paired per-(region, seed) difference.

Metric	Nominal		Turbulence		p
	LQR	H_∞	LQR	H_∞	
Cross-track RMS (cm)	—	—	— $\pm$ —	— $\pm$ —	0.0—
Cross-track p95 (cm)	—	—	— $\pm$ —	— $\pm$ —	0.0—
Cross-track max (cm)	—	—	— $\pm$ —	— $\pm$ —	0.0—
Longitudinal lag (cm)	—	—	— $\pm$ —	— $\pm$ —	0.0—
Altitude RMS (cm)	—	—	— $\pm$ —	— $\pm$ —	0.0—
Roll p95, sweeping (deg)	—	—	— $\pm$ —	— $\pm$ —	0.0—
Pitch p95, sweeping (deg)	—	—	— $\pm$ —	— $\pm$ —	0.0—
Torque integral J_τ	—	—	— $\pm$ —	— $\pm$ —	0.0—
RPM RMS	—	—	— $\pm$ —	— $\pm$ —	0.0—

[TODO: figures/fig5_crosstrack_spread.png — box plot cross-track RMS LQR vs H_∞ lintas 5 seed, per region. Inilah yang membenarkan pengulangan berseed; tanpa gambar ini angka $\pm$ sd tidak terbaca.]

FIGURE 9. Spread of cross-track RMS error across gust seeds. Boxes span the inter-quartile range over the five seeds; whiskers give the full range.

TABLE 8. Degeneration of the H_∞ synthesis at large γ . Gains are those returned by Eq. (32); the last column is the relative deviation from the LQR gain of Eq. (31) for the same subsystem.

Subsystem	γ	K_p	Dev. from LQR (%)
<i>Earlier setting, far above $\gamma_{\min}$</i>			
Outer position	80	—	—
Inner attitude	64	—	—
Yaw	64	—	—
<i>Operating setting, $\gamma = 1.3 \gamma_{\min}$</i>			
Outer position	3.30	—	—
Inner attitude	3.31	—	—
Yaw	11.73	—	—

Two quantities carry the argument. The path-length increase is the price of routing; the coverage loss $\Delta\eta$ against the nominal condition is what that price buys. Because the keep-out radius is smaller than the sensing radius, Eq. (5), the annulus around each cylinder is mapped during the detour, so $\Delta\eta$ is expected to stay small even as the path lengthens. The clearance and feasibility rows test the barrier itself: a $P(\text{tier} > 0)$ that stays near zero means the quadratic program was solvable at its nominal tier almost everywhere, so the relaxation ladder is a safety net, not an operating mode.

Because the map starts empty, the detours in this condition can only be planned once a cylinder has been found. Table 9 reports how well the swarm discovered the field it was flying through, and Fig. 10 shows the discovery curve against mission time. **[TODO: isi dari log ObstacleMap: bandingkan track terkonfirmasi terhadap O^{true} , catat jarak pertama terdeteksi, dan hitung berapa baris sapuan yang sempat direncanakan sebelum rintangannya diketahui.]**

Fault-Tolerant Recovery after Agent Loss

TABLE 9. Online obstacle discovery. Nine cylinders are present in each region; the planner sees only confirmed tracks.

Quantity	L-shape	U-shape	Plus
Cylinders confirmed (of 9)	-	-	-
False tracks confirmed	-	-	-
Time to first confirmation (s)	—	—	—
Time to last confirmation (s)	—	—	—
Detection range, median (m)	—	—	—
Rows replanned after discovery	-	-	-

[TODO: figures/fig6_discovery.png — jumlah silinder terkonfirmasi terhadap waktu misi untuk ketiga region, garis putus-putus pada 9 (jumlah sebenarnya), dan penanda saat re-plan dipicu]

FIGURE 10. Online obstacle discovery against mission time. The dashed line marks the nine cylinders actually present; the gap between the curve and that line is the part of the field the planner was still unaware of, during which safety rested on the barrier layer alone.

TABLE 10. Coverage and safety under turbulence with static obstacles; mean $\pm$ sd over $N_r = 5$ seeds. Δ columns are measured against the nominal condition of Table 5.

Metric	L-shape		U-shape		Plus	
	LQR	H_∞	LQR	H_∞	LQR	H_∞
Final coverage η (%)	—	—	—	—	—	—
Coverage loss $\Delta\eta$ (pp)	—	—	—	—	—	—
Path length increase (%)	—	—	—	—	—	—
Min. obstacle clearance (m)	—	—	—	—	—	—
$d_{\min}$ inter-agent (m)	—	—	—	—	—	—
$P(\text{tier} > 0)$ (%)	—	—	—	—	—	—
Collisions	-	-	-	-	-	-

[TODO: figures/fig3_scheme4_tracks.png — lintasan kondisi 3 dan 4 untuk ketiga region dan kedua kontroler, dengan sembilan silinder digambar dan titik kegagalan ditandai]

FIGURE 11. Flown tracks under turbulence and obstacles. Cylinders are drawn to their keep-out radius; crosses mark the two agent failures of condition 4.

TABLE 11. Fault-tolerant recovery: detection and coverage outcome. Mean $\pm$ sd over $N_r = 3$ seeds; $\Delta\eta$ against condition 3.

Metric	L	U	Plus
Δt_{det} , fail 1 (s)	--	--	--
Δt_{det} , fail 2 (s)	--	--	--
Δt_{rec} , fail 1 (s)	--	--	--
Δt_{rec} , fail 2 (s)	--	--	--
Final coverage η (%)	--	--	--
Coverage loss $\Delta\eta$ (pp)	--	--	--
Duration increase (%)	--	--	--

TABLE 12. Fault-tolerant recovery: reallocation load and safety maintained during the recovery phase.

Metric	L	U	Plus
Recovery rows, fail 1	-	-	-
Recovery rows, fail 2	-	-	-
Rows rescued as orphans	-	-	-
Helper agents allocated	-	-	-
d_{min} , recovery (m)	--	--	--
Min. obstacle clear. (m)	--	--	--
Collisions	-	-	-

[TODO: figures/fig4_load_ladder.png — coverage terhadap waktu, tiga kurva bertumpuk per region: nominal, turbulensi+rintangan, dan +kehilangan agen, dengan penanda vertikal di $t_{f,1}$ dan $t_{f,2}$]

FIGURE 12. Coverage against mission time along the load ladder. The vertical gap between the curves with and without agent loss at mission end is $\Delta\eta$, the coverage attributable to agent loss alone; the vertical lines mark the two failures.

TABLE 13. Constraint-set size actually solved, sampled over all ticks. Feeds the $O(m^2)$ cost of Table 14.

Row class	Median	Max
Static obstacles	-	-
Inter-agent (V2V)	-	-
Walls + stop planes	-	-
Speed + rate box	16	16
Total m	-	-

TABLE 14. CBF-QP solve time per agent per tick, from a standalone benchmark of the solver over representative constraint sets.

Quantity	Median	p99
Constraint rows m	-	-
Solve time (μs)	--	--
Per-tick budget for 7 agents (μs)	--	--
Fraction of the -- ms control period (%)	--	--

Condition 4 repeats condition 3 with two sequential failures, so the difference between Tables 11–12 and Table 10 isolates the cost of losing agents from the cost of the obstacle field. [TODO: isi dari kondisi 4 (18 misi)]

The safety rows matter as much as the coverage rows. Reallocation moves helpers into cells they did not plan for, while the wind is still acting and the cylinders are still present, which is the most demanding load the barrier layer sees anywhere in this study. Separation holding through that phase is the strongest available evidence that the class- $\mathcal{K}$ function of Eq. (17) is correctly matched to the vehicle.

Computational Cost of the Safety Filter

Actuator effort is reported alongside the tracking metrics in Table 7, since it is the price of that tracking. What remains is computation, which determines whether the safety filter is admissible at the control rate. Table 13 gives the size of the constraint set actually solved and Table 14 the resulting solve time. [TODO: isi kedua tabel; biaya komputasi dari micro-benchmark solver, bukan dari misi]

The solver is exact rather than iterative, so the p99 figure is a genuine worst case over the enumerated candidate set of Eq. (25) and not a convergence tail. [TODO: setelah terisi, nyatakan margin real-time-nya terhadap periode kendali.]

Comparison with Reported Results

Re-implementation of the baselines was out of scope, so Table 15 positions the present results against the figures their authors report. The comparison is indicative, not controlled: domains, agent counts, and sensing models differ,

TABLE 15. Indicative positioning against reported results. Entries are the values stated by the cited authors under their own conditions; “n/r” marks a quantity those authors do not report.

Work	Domain	Agents	Coverage	Full 6-DOF	Fault tol.
Cortés <i>et al.</i> [5]	convex	–	n/r	no	no
Breitenmoser <i>et al.</i> [8]	non-convex	–	–	no	no
Lee and Lee [9]	non-convex	–	–	no	–
Cao <i>et al.</i> [10]	non-convex	–	–	no	no
This work	non-convex	7	–	yes	yes

and only [8] and [9] evaluate non-convex domains at all. No claim of superiority is made on its basis.

[TODO: isi kolom dari angka yang DILAPORKAN di paper masing-masing, jangan dikira-kira; tandai "n/r" untuk yang tidak dilaporkan penulisnya.]

LIMITATIONS

State feedback is ideal. Agents receive ground-truth odometry; there is no estimator noise, bias, or latency. The absolute tracking errors reported here are therefore a lower bound for an equivalent flight system.

Coverage is geometric, not radiometric. As set out with Eq. (5), η counts area observed within the sensing radius at the required GSD. It does not model illumination, motion blur, feature matching, or bundle-adjustment residuals, so a mission reported here as fully covered is one that satisfies the geometric precondition for mapping, not one whose reconstructed map has been validated. Closing that gap requires a photogrammetric pipeline in the loop and is the natural next step for the payload side of this work.

Moving obstacles are excluded. The load ladder stops at static cylinders. The constraint rows of Eq. (20) already carry the obstacle velocity term, so the formulation extends to translating obstacles, but the resulting behaviour is not yet stable enough to report and no moving-obstacle results are claimed.

Fault tolerance is demonstrated on two sequential losses only. The scenario terminates at most two of seven agents; behaviour under larger or simultaneous losses, and under communication loss without vehicle loss, is untested.

Execution is centralised. The formulation is per agent, but the seven agents are solved inside one process so that the reciprocal split of Eq. (26) is exact. Distributing them across vehicles would replace each neighbour’s current weight w_j with a value received over a link. Because w_j changes whenever agent j changes mission state, a stale value breaks $\lambda_{ij} + \lambda_{ji} = 1$ and with it the joint condition $\dot{h} \geq -\varphi(h)$: if both agents happen to believe they are the protected one, neither gives way. A distributed deployment would therefore need either a conservative fallback, $\lambda_{ij} = 1$ whenever the last update from j is older than a bounded age, or a fixed symmetric split at the cost of the mission-state weighting. Neither variant is evaluated here. The partitioning layer is less affected, since decentralised centroidal Voronoi coverage is established [5], whereas FT-CC would additionally require the survivors to agree on which agent has failed and which rows each will take.

Baselines are not re-implemented. Table 15 positions this work against published figures obtained under different domains, agent counts, and sensing models. It is not a controlled comparison, and no claim of superiority over those methods is made on its basis.

No physical flight. Ground effect, inter-agent rotor wash, and airframe flexibility are absent from the model and untested.

CONCLUSION

An end-to-end three-layer architecture for geodetic mapping of non-convex regions by a quadrotor swarm was presented and evaluated on full six-degree-of-freedom dynamics. The high-level layer partitions an arbitrary polygon by centroidal Voronoi tessellation, Eqs. (1)–(3), and sweeps each cell with a boustrophedon path whose extremal rows follow the cell boundary, Eq. (7), removing the unswept wedges a fixed-pitch sweep leaves at tapered corners. The mid-level layer resolves all safety constraints in one quadratic program per agent per tick, Eq. (24), whose class- $\mathcal{K}$ function, Eq. (17), is derived from the identified closed-loop plant, not tuned by hand. The low-level layer compares LQR and H_∞ synthesis, Eqs. (31)–(34), on an identical cascade.

Evaluated along a load ladder that adds one stressor at a time over three non-convex regions, the swarm held

XX.X-XX.X% coverage with zero inter-agent collisions at every rung, and the quadratic program remained feasible at its nominal tier in essentially every tick. Adding nine obstacles per region lengthened paths by **XX%** while costing only **X.X** percentage points of coverage, because the detour radius stays inside the sensing footprint. Losing two of seven agents on top of that cost a further **X.X** points, with the first helper sweeping a recovery row **XX** s after the failure and separation maintained throughout the reallocation. Under Dryden turbulence H_∞ reduced cross-track RMS error by **XX%** relative to LQR at **X.Xx** the control effort. A methodological result of independent interest is that the H_∞ attenuation bound must be chosen near its feasibility limit, Eq. (35); at large nominal values the Riccati equation degenerates and the robust controller silently becomes a disguised LQR.

Future work follows from the limitations: completing the obstacle scenarios, extending fault tolerance to larger simultaneous losses and to communication loss without vehicle loss, placing a state estimator in the loop, and carrying the architecture to outdoor flight and heterogeneous teams.

ACKNOWLEDGMENTS

The authors thank the Center for Instrumentation Technology and Automation (CITA), Institut Teknologi Bandung, for supporting the conference registration and publication fees for this paper. CITA provided no research funding, and had no role in the design of the study, the simulations, the analysis, or the decision to publish.

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